

Gajendra Saraswat

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with **3+ years** (Aug 2023 – Present) building and operating **enterprise-scale distributed systems** across **Java/Spring Boot/Angular** (Contract Platform) and **monolithic ASP.NET/IIS** (Regulatory Platform) at BNY Mellon. Proven in designing highly available microservices, engineering disaster recovery for 25+ applications, Kafka multi-cluster resiliency, and zero-downtime releases. Delivered **36% faster contract processing** (Contract Platform), **99% faster page loads** (7 min -> 5 sec for 18.8 MB, Regulatory Platform), **4,000+ security issues remediated**, and **300+ scenarios / 95% automated coverage** evolved to fully automated Jira/MCP regression. Strong in RBAC, secure REST APIs, Windows Server/IIS, F5/Cumulus, VMAD - Vulnerability Management, SiteMinder, and regulated-environment constraints. Daily user of **AI-assisted development (Windsurf, Devin)** for code generation, refactoring, and documentation with governance.

TECHNICAL SKILLS

- **Languages:** Java (8, 17), Python, C++, SQL
- **Backend & Frameworks:** Spring Boot, Spring MVC, Spring Data JPA / Hibernate, RESTful APIs, Microservices, Apache Kafka, ASP.NET (legacy), API Gateway, API Manager
- **Architecture & Design:** Distributed Systems, Microservices, Monolithic (Regulatory Platform), API Design, System Design, Scalability, Resilience, Fault Tolerance
- **Datastores:** MySQL, PostgreSQL, Redis, Vertica, DB2, Informatica (Regulatory Platform, ETL - platform-level), Relational Schema Design, Query Optimization, Caching
- **DevOps & Platform:** Docker, Jenkins, Git, GitLab CI/CD, Maven, JFrog Artifactory, Ansible (Regulatory Platform), Google App Engine (Contract Platform), F5 BIG-IP, Cumulus, RDP, IIS, Windows Server 2016/2022
- **Testing & Quality:** JUnit, Mockito, Integration Testing, Selenium, Gherkin, Cucumber, SonarQube, Code Coverage, MCP, Jira, Confluence
- **Observability:** Grafana, Splunk, Orion, PVT, Structured Logging, Performance Monitoring
- **Security & Access Control:** Authentication, Authorization, RBAC/Entitlements, Secure API Development, VMAD - Vulnerability Management, SiteMinder, CyberArk PAM, Active Directory, Secret Scanning
- **AI-assisted Development:** Windsurf, Devin, Prompt Engineering, RAG, Agent Design, Code Generation, Refactoring, Debugging, Documentation
- **Ways of Working:** Agile/Scrum, RFCs, Design/Code Reviews, Production Support (L2/L3), BAU, Incident Response, Runbooks

EXPERIENCE

Bank of New York Mellon

Chennai, India

Software Engineer (Backend)

Aug 2023 – Present

- **Microservices & Contract Platforms (Contract Platform – Java/Angular):** Own backend services for contract maintenance and reconciliation; build **Spring Boot** microservices handling high-volume financial workloads with **@Async** and **@Transactional** for concurrency and data integrity. Expose secure REST APIs and **Apache Kafka**-backed async integration via **API Gateway / API Manager** (throttling, rate limiting, sandbox vs prod URLs); deploy Contract Platform via **Google App Engine**, Docker, Jenkins, GitLab CI/CD.
- **Performance Engineering – Dual-Track Wins: Contract Platform:** Optimized contract upload pipelines via parallelization, batching, and tuned transaction boundaries -> **36% reduction** in end-to-end processing time and predictable SLAs; orchestrated client-facing URI migration ***.bnymellon.* -> *.bny.*** across APIs, F5, and assembly configs with zero perceived downtime. **Regulatory Platform (ASP.NET Legacy):** Reduced 18.8 MB heavy-page load **7 min -> 1 min (86% reduction)** by enabling IIS static/dynamic compression via RDP, then -> **5 sec (99% total reduction)** via **application-wide server-side pagination** rolled to all pages (not one-off).
- **Disaster Recovery & Kafka Resiliency (Contract Platform):** Engineered disaster recovery workflows for **25+ enterprise applications** across multi-cluster infrastructure, enabling automated failover and consistent runbooks. Validated **Kafka backup cluster resiliency (8 topics, 2 clusters):** ensured automatic switchover from Cluster 1 to Cluster 2, verified offsets (earliest/latest), consumer lag, and **latest vs earliest** strategies in lower environments before production failover.
- **Infrastructure Migration & Reliability (Regulatory Platform – Windows/IIS):** Led Windows Server **2016 -> 2022** migration (QA: 3 -> 2 servers, PROD: 3 -> 3 servers) across QA/UAT/PROD for regulatory reporting platform supporting cross-border holdings (backed by DB2, Informatica ETL at platform level). Deployed via **Ansible** (Regulatory Platform), updated **F5 BIG-IP** and **Cumulus** load balancers (Cumulus URLs), validated per-server via isolated Chrome direct-URL sessions testing all functionalities; monitored with **Grafana, Splunk, Orion, PVT**; smoothly decommissioned old servers with zero downtime.
- **Security, Entitlements & AI Governance:** Led remediation of **4,000+ security findings** in 3 years across distributed modules (incl. **Regulatory Platform: 28 total – 5 Critical, 8 High, remainder Medium/Low**) hardening login/auth (validated via SonarQube, VMAD - Vulnerability Management) and aligning to organizational standards. Implemented authentication/authorization and granular **RBAC entitlements** (Approver, View, Add) for contract/product data enforcing least-privilege; **go-to person for entitlements** – create entitlements, map applications to level-based accesses, fulfill prod access via **Access Hub** request forms and directly run internal provisioning for lower environments. Reviewed **Devin Chain-of-Thought credential-exposure report** scanning chats, GitLab commits/pipelines; remediated exposed secrets, rotated credentials, and authored preventive guidelines to block recurrence.
- **Production Incident Response & Adaptability:** Diagnosed **P1 MIME-type outage** after Edge (then Chrome) strict enforcement broke Regulatory Platform: global **web.config** rule wrapping all **.axd/.css/.js/.html** as **text/html** violated new browser expectations; coordinated with **SiteMinder** team, removed rule and restored service. Adapted quickly to **new ASP.NET/IIS monolithic stack**. Separately resolved regulatory **report-generation failure** before deadline, preventing client-facing downtime and regulatory impact.

- **Test Automation, Delivery & BAU:** Built and expanded **NextGen automation suite (Selenium, Gherkin, Cucumber)** from **300+ scenarios / 95% coverage** to **fully automated regression**: scripts auto-generate test cases and reports, auto-push to **Jira via MCP (Model Context Protocol)**, auto-update release metrics (% automated per release). Drive fixes via **RFCs, design/implementation reviews, runbooks**. Own **BAU, L2/L3 maintenance, vulnerability triage**, and cross-functional collaboration with SMEs/platform/security.
- **Developer Productivity & AI Tooling:** Daily use of **Windsurf and Devin** for prompt-driven code generation, refactoring, debugging, test and documentation drafting, and RFC authoring; integrate AI into backend workflows using patterns from **AI Pioneer** (RAG, Agent Design, Nuggets, workflow orchestration) while enforcing secure, governed use.

SELECTED PROJECTS

- **High-throughput Contract Processing Service** *Mar 2025 – Banking Domain (Contract Platform)*
Enhanced Spring Boot contract processing service for higher volume and stricter SLAs. Refactored critical workflows with **@Async/@Transactional**, validation, idempotency, and robust error handling -> contributed to **36%** reduction; expanded JUnit/Mockito/API test coverage improving deployment confidence.
- **Credit Card Fraud Detection System** *Sep 2020 – Final Year Project (Backend & ML) [GitHub 16 stars / 16 forks]*
Built **Python, Flask** backend exposing RESTful APIs for fraud detection with scalable architecture and efficient data access. Implemented request validation, rate limiting, query optimization, and caching; containerized with **Docker** and deployed on **Heroku** with structured logging/monitoring. Tech: Python, Flask, RESTful APIs, Docker, Heroku.

EDUCATION

- **National Institute of Technology, Warangal** Warangal, India
Master of Technology, Computer Science; CGPA: 7.13 *Sep 2021 – Jun 2023*
- **MBM Engineering College, Jodhpur** Jodhpur, India
Bachelor of Engineering, Computer Science; CGPA: 7.48 *Aug 2016 – Oct 2020*

PUBLICATION

- **Autoencoder-Huffman Ensemble Model for Image Compression** *Dec 2024*
Springer, Singapore – link.springer.com/chapter/10.1007/978-981-97-6714-4_18
Proposed two-stage autoencoder-Huffman approach achieving **5.5x–24.2x** compression ratios on MNIST, CIFAR-10, Fashion MNIST with minimal information loss.

HONORS & AWARDS

- **GATE 2021:** Secured **99.1 percentile (AIR 943)** in Graduate Aptitude Test in Engineering (Computer Science)
- **Kaggle SIIM-ISIC Melanoma Classification:** Ranked in **top 19% globally** in large-scale medical imaging competition

CERTIFICATIONS

- **AI Pioneer – Proficiency Level 3:** BNY Internal AI Learning Program – RAG, Agent Design, Prompt Engineering, Nuggets, Workflow Orchestration, AI Governance
- **Machine Learning** – Coursera
- **Deep Learning Specialization** – Coursera
- **Introduction to Data Science in Python** – Coursera
- **Statistical Thinking in Python (Part 1)** – DataCamp
- **Mathematics for Machine Learning** – Coursera

Keywords for ATS (visible): Java, Spring Boot, Microservices, REST APIs, Apache Kafka, MySQL, PostgreSQL, Redis, Vertica, Docker, Jenkins, GitLab CI/CD, Ansible (Regulatory Platform), Google App Engine (Contract Platform), F5, IIS, Windows Server, ASP.NET, VMAD - Vulnerability Management, SiteMinder, CyberArk, Grafana, Splunk, JUnit, Selenium, Cucumber, Windsurf, Devin, RAG